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|  <b>sentera</b> | Aug 28, 2025      |
| <b>6X Sensor - Programming Instructions</b>                                                      | Doc # 27075 Rev-C |

## 1 Hardware

- Windows Computer
- USB-C to USB Cable
- 24182 – Cable Assembly, Power, Focusing
- 24184 – Cable Assembly, Focusing Interface, 6X
- FLIR Boson VPC Adapter

## 2 Software/Firmware

- 6X-21214\_Programming
- 6XT-21216\_Programming
- FLIR Boson App

### 2.1 FLIR Boson App Install (21216 Only)

If not already installed, the following steps need to be completed to install the FLIR Boson App on the user's PC.

1. Copy to a local directory the installation file ***bosonAPP\_v3.0.0\_x64drivers.msi***:

\\172.25.0.194\Data\Part Database (things we sell)\SW-xxxx-xxxx -- 6x Firmware\FLIR Boson\GUI

2. Run the ***bosonAPP\_v3.0.0\_x64drivers.msi*** installation file.
3. Follow the wizard instructions to complete the installation.

## 3 Sensor programming

This section covers the programming steps for both the thermal and multispectral variants of the 6X sensor. Files specific to a variant are included in their applicable sub-folder:

- 6X: ***6X-21214\_Programming***
- 6XT: ***6XT-21216\_Programming***

Section 3.1 below pertains only to the thermal 6X and should be done before the sensor is installed into the lens mount.

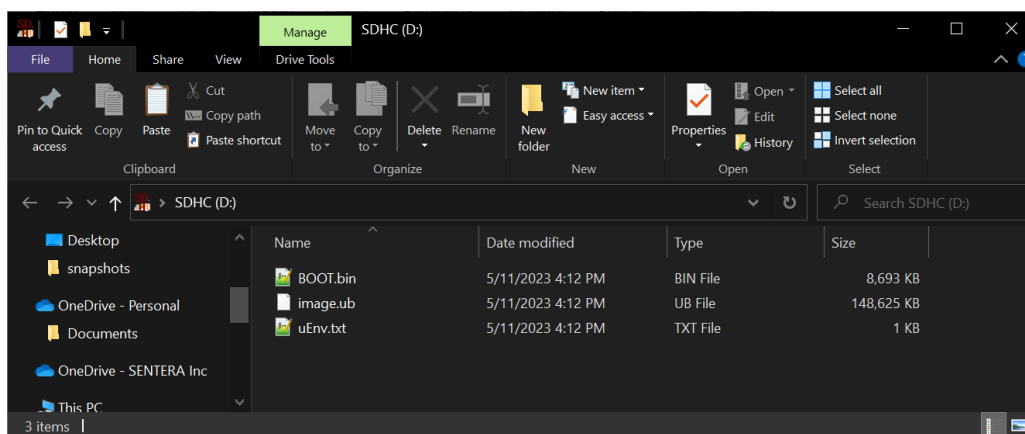
### 3.1 Thermal Sensor Configuration (21216 Only)

1. Connect the FLIR Boson VPC adapter to the back of the thermal sensor.
2. Connect the USB cable between the PC and the VPC adapter.
3. Open the FLIR Boson App.
4. In the lower right corner of the window, click the **"Port"** dropdown and select the COM port for the thermal sensor.
  - a. If the thermal sensor is the only USB device used, there will only be one COM port.
  - b. The correct COM port can also be viewed in the device manager.
5. The **"Boson Link"** status should show connected.
6. Click the **"Image Appearance"** button on the left column of options.

7. Click on the **“Analog/CMOS”** video controls section.
  - a. Under the video section, click the **“CMOS”** checkbox.
  - b. Under the configure section, set video source to **“TLinear”**. The Output should be **“TLinear”**.
  - c. Under the CMOS & USB telemetry section, turn **“CMOS Enable”** on.
  - d. Under the CMOS & USB telemetry section, set the **“Mode”** drop down to **“Header”**.
8. Click on the **“System”** button on the column on the left column of options.
  - a. Under the **“Configuration Controls”** section, click **“Save Power-On Defaults”**.
9. Disconnect the sensor and close the GUI.
10. To verify that the correct changes were made, reconnect the sensor, reopen the FLIR Boson app, and ensure that all the settings changes remained.

## 3.2 Programming – Sensor Assembly

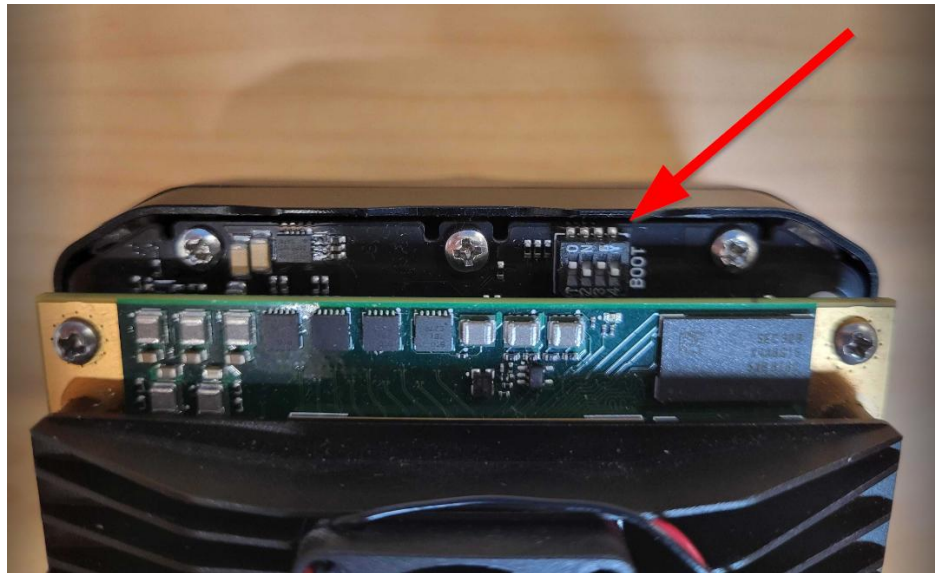
1. Remove the microSD card from the camera and put it into the computer.
2. Copy the contents of folder ***firmware - sdcard*** to the microSD card.



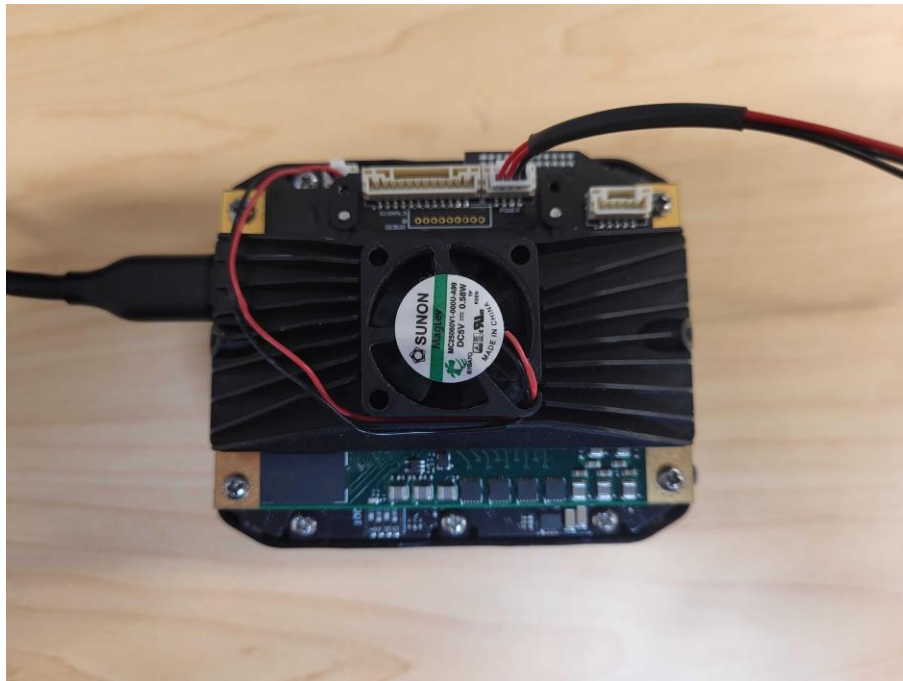
3. Eject the microSD card from the computer and put it inside the camera.
4. Remove the rear cover of the camera (if present).
5. Set the dip switches present on the Imager Baseboard CCA to the following positions:

| Dip Switch | Position |
|------------|----------|
| 1          | ON       |
| 2          | OFF      |
| 3          | ON       |
| 4          | OFF      |

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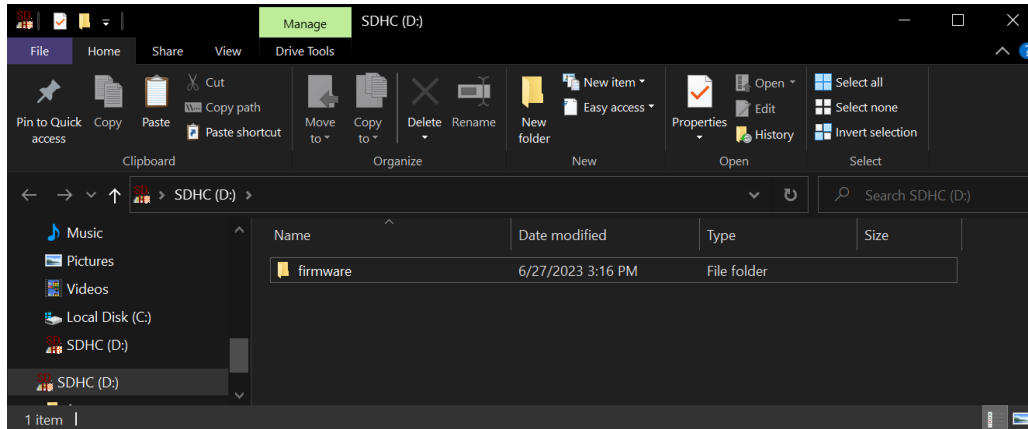
6. Connect the *Power cable (24182 & 24184)* between an outlet and the camera.
7. Connect the *USB-C to USB cable* between the PC and the camera.



8. Navigate to 192.168.42.1 on a web browser.
9. Navigate to the "Update Firmware" tab on the left side of the webpage.
10. Click in the "Firmware Update" field and select the firmware update file (where "X.X.X" is the release version) from the *firmware – factory update* folder:
  - a. 6X: ***6x-factory-update\_X.X.X-21214.swu***

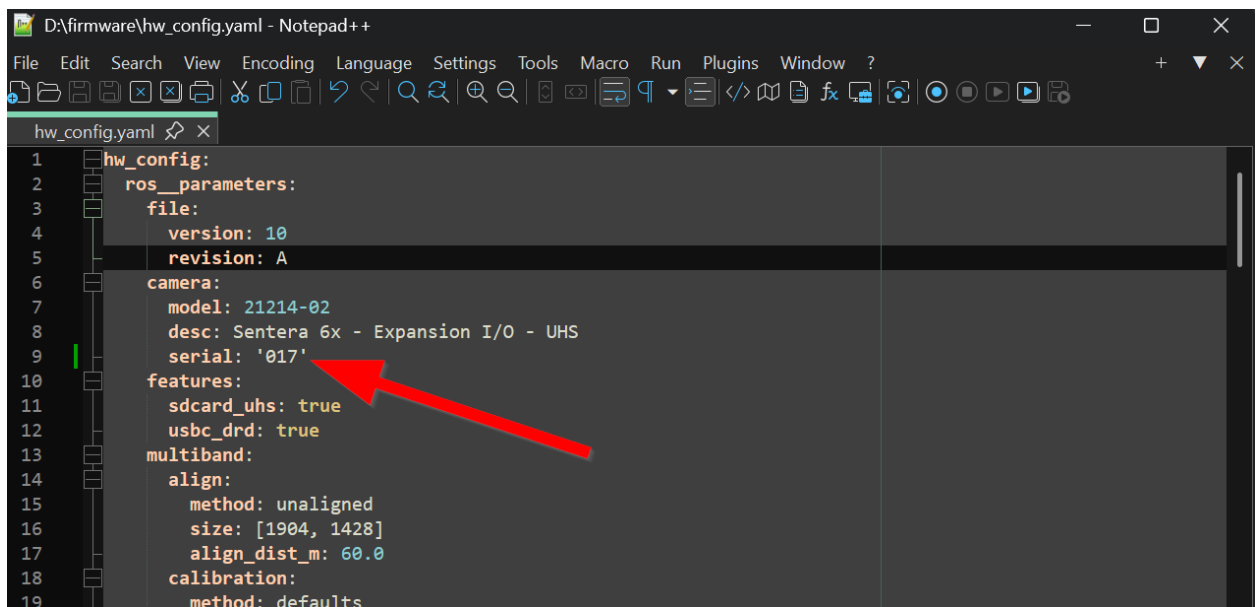
b. 6XT: **6x-thermal-factory-update\_X.X.X-21216.swu**

11. Once the firmware update is completed (approx. 5 minutes) power down the camera.
12. Remove the microSD card from the camera and place it in the computer.
13. Delete all files from the microSD card.
14. Copy folder **configs/2121X-XX/firmware** to the microSD card. Select the **2121X-XX** folder based on what camera model is being programmed.



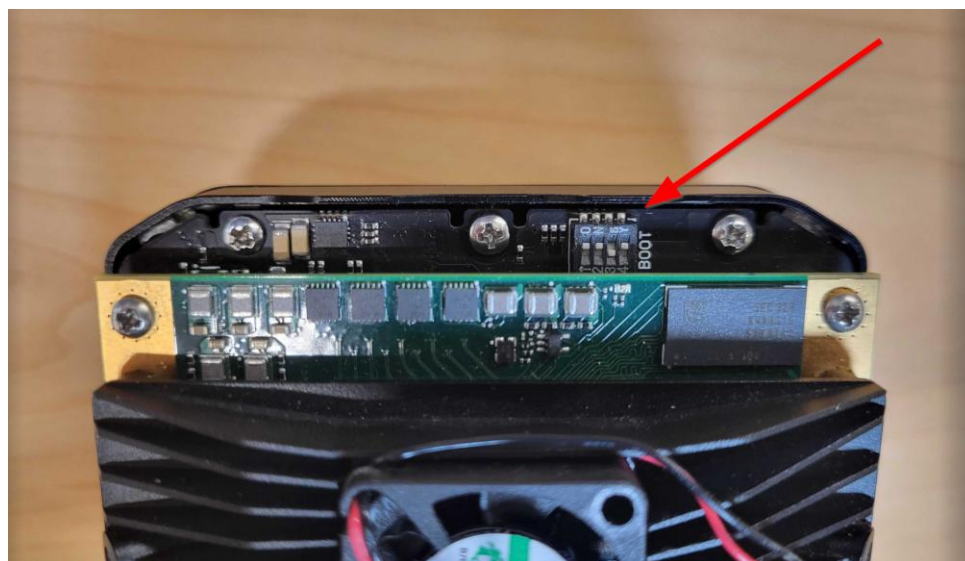
*Note: The OEM – Sentera GPS configuration is applied as the initial configuration so that camera operation and focusing can be accomplished without being connected to a gimbal.*

15. On the microSD card, open file **firmware/hw\_config.yaml**, and update the serial number to the serial number of the camera (surrounded by single quotes).

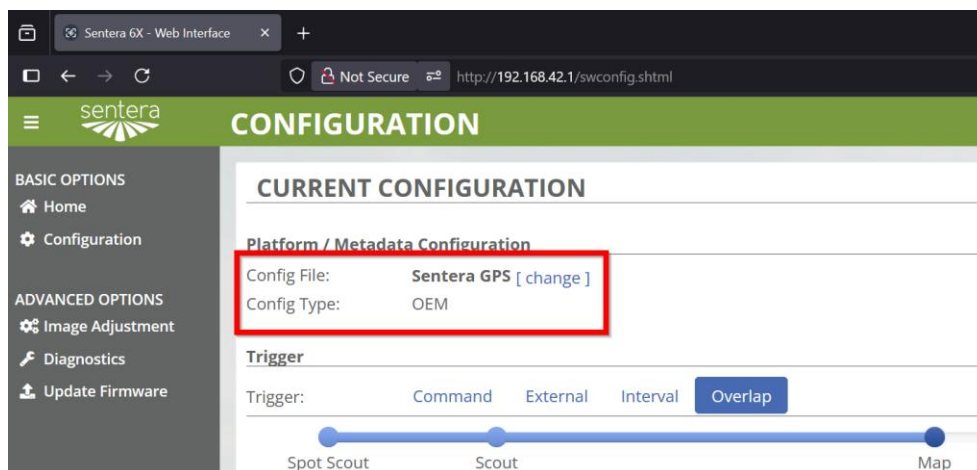


16. Eject the microSD card from the computer and place it in the camera.
17. Set the dip switches present on the Imager Baseboard CCA to the following positions:

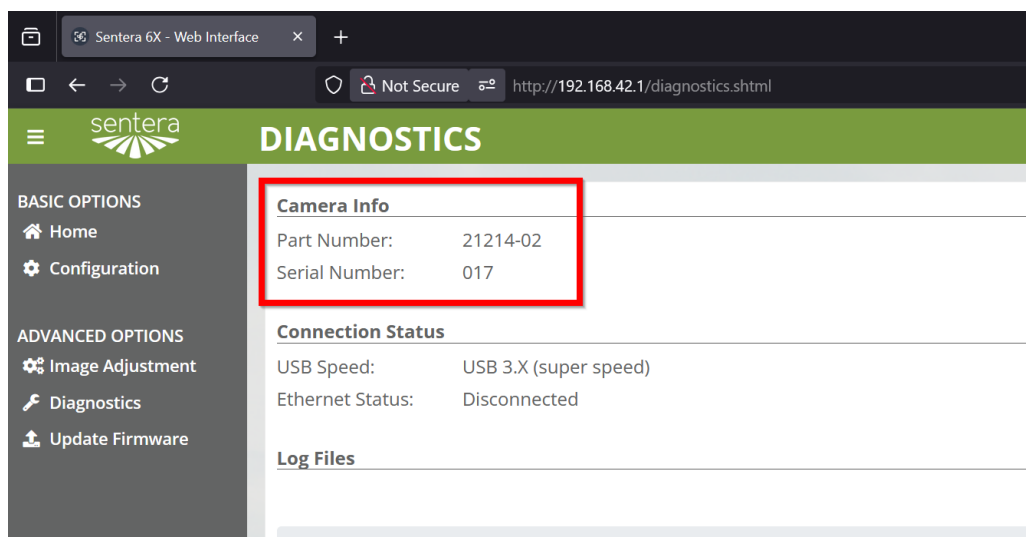
| Dip Switch | Position |
|------------|----------|
| 1          | ON       |
| 2          | ON       |
| 3          | OFF      |
| 4          | ON       |



18. Re-install the rear cover of the camera.
19. Power on the camera.
20. Navigate to 192.168.42.1 on a web browser.
21. Navigate to the "Configuration" tab on the left side of the webpage.
22. Verify that the "OEM – Sentera GPS" configuration is being used.



23. Navigate to the “Diagnostics” tab on the left side of the webpage.
24. Verify that the Part Number and Serial Number are what is expected for the camera.



25. Programming is complete!